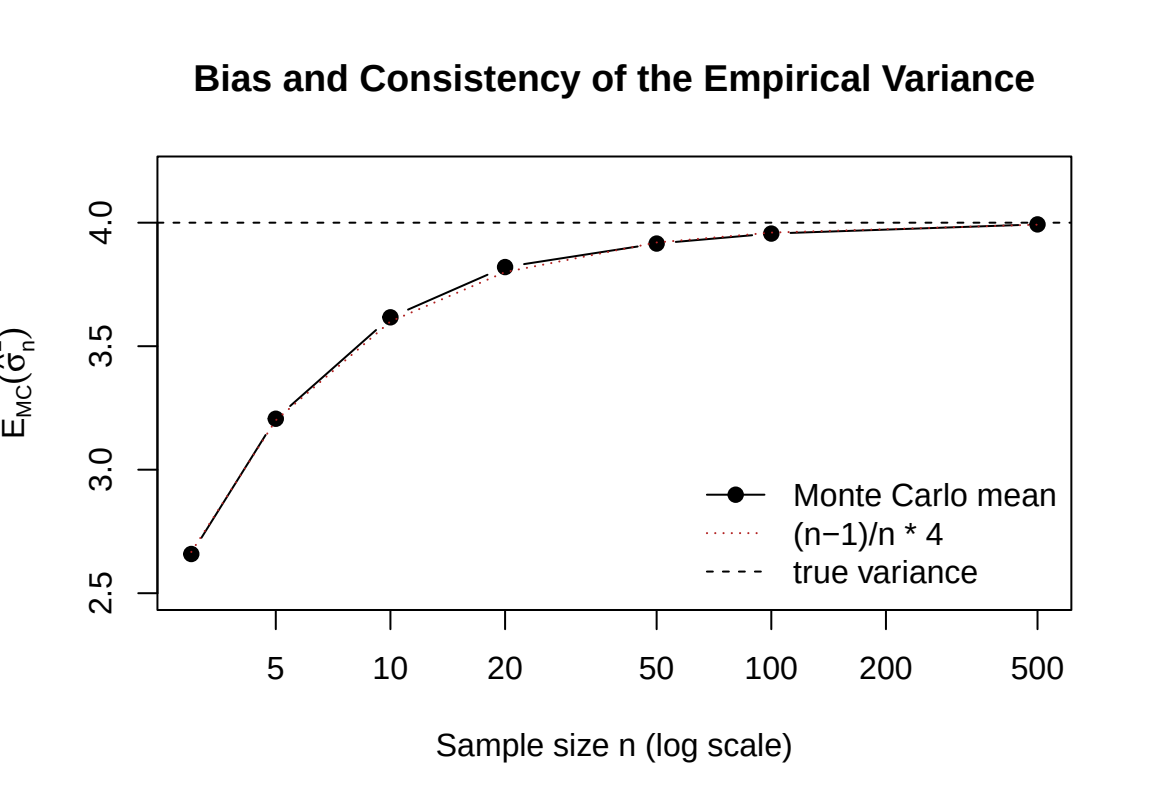
Bias and Consistency of the Empirical Variance
$E_{MC}(\hat{\sigma}_n^2)$
4.0
3.5
3.0
2.5
Monte Carlo mean
(n−1)/n * 4
true variance
5
10
20
50
100
200
500
Sample size n (log scale)